

SURAJ CHOUHAN

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PROFESSIONAL SUMMARY

AI/ML Engineer focused on building production-grade Generative AI and intelligent automation systems. Expertise in LLM applications, RAG architectures, NLP, and computer vision using LangChain, OpenAI, Gemini, PyTorch, and FastAPI. Published researcher in medical AI with strong experience designing scalable AI solutions and modern MLOps workflows.

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, CNNs, Embeddings, Diffusion Models
GenAI: LLMs, RAG, LangChain, Agents, Prompt Engineering, Vector DBs (FAISS, Pinecone, Milvus), OpenAI GPT-4o, Groq, Gemini, Hugging Face
Frameworks: PyTorch, TensorFlow, Scikit-learn, Transformers, FastAPI, Flask, OpenCV, CLIP
Cloud & MLOps: AWS (EC2, S3, RDS), Azure, Pulumi, Docker, MLflow, Airflow, CI/CD
Databases: PostgreSQL, MySQL, FAISS, Pinecone
Programming: Python, SQL, JavaScript, HTML, CSS

PROFESSIONAL EXPERIENCE

Gems Essence Infotech Pvt. Ltd. Indore, India
Software Developer (AI/ML) August 2025 – Present

- Spearheaded development of GenAI & Agentic AI systems using LangChain and GPT-4o, delivering FastAPI-based inference APIs handling 5K+ queries/day while designing intelligent document-processing pipelines achieving 92% classification accuracy for legal and financial datasets.
- Streamlined MLOps workflows with Docker and MLflow, enabling reproducible training pipelines and accelerating model deployment cycles by 45%.
Trade Intelligence — Agentic AI & Financial Data Automation
- Designed a multi-agent Text-to-SQL analytics platform on PostgreSQL using MCP for context orchestration, enabling natural-language financial insights and automated reporting agents that reduced manual reporting workload by 70%.
AI-Medi Bot — Multimodal Generative AI Medical Assistant
- Created a multimodal diagnostic assistant using FastAPI and Gemini 2.5 Flash supporting X-ray, CT, and dermatology image analysis with explainable heatmaps and multilingual voice interaction across 8+ languages while aligning with HIPAA/GDPR compliance standards.

Techinfini Solutions Pvt. Ltd. Indore, India
Associate Software Engineer August 2023 – August 2025

- Delivered AI-powered chatbot and sentiment analysis platforms using OpenAI, Groq, and Hugging Face models serving 10K+ monthly users while constructing CNN-based medical image classifiers and improving Shopify GraphQL API performance by 35%.
- Automated product feed pipelines for Google Merchant Center and Facebook Ads, eliminating 80% of manual catalog processing and accelerating campaign launch cycles.
GenAI-Powered Ad Generator — LLM & Marketing Automation
- Built a generative marketing automation platform leveraging Gemini and Stable Diffusion to generate high-quality ad creatives from product URLs, reducing marketing content creation time by 60%.
Safe Chat Guard — Real-Time AI Content Security
- Deployed a real-time AI moderation framework using OpenAI Moderation API and WebSockets enabling automated toxicity detection and compliance monitoring dashboards.
Fashion Search Bot — Image-Based Recommendation Engine
- Constructed a visual search system utilizing CLIP embeddings and cosine similarity to enable real-time fashion product matching across 100K+ product images.

Forge Foundation Remote
Software Engineering Virtual Intern March 2022 – April 2022

- Completed Walmart Global Tech Advanced Software Engineering simulation focused on scalable system architecture, database design, and enterprise development practices.

RESEARCH

Multimodal Fusion of CT and MRI for Liver Tumor Segmentation Using Attention-Based CNNs 
Journal of Neonatal Surgery, Vol. 14 No. 31S (2025) — Enhanced liver tumor detection via multimodal image fusion and attention-based deep learning improving diagnostic precision in medical imaging systems.

EDUCATION & CERTIFICATIONS

Medi-Caps University, Indore, India July 2023 – May 2025
Master of Technology (M.Tech) in Artificial Intelligence GPA: 9.2 / 10
Indore Institute of Science & Technology, Indore, India Aug 2019 – Jun 2023
Bachelor of Technology (B.Tech) in Computer Science GPA: 8.59 / 10

Certifications: AWS Machine Learning Specialty, AWS Cloud Foundations, IBM Web Development.